

Traveling Wave Linear Accelerator With RF Power Flow Outside of Accelerating Cavities

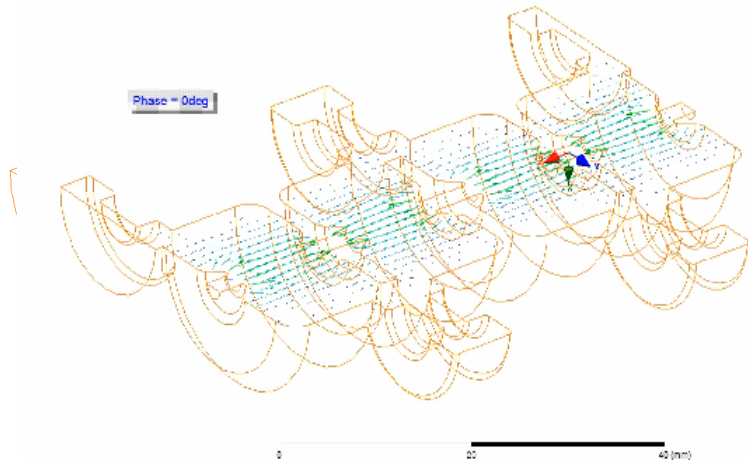
V. A. Dolgashev, SLAC

LINAC 2016, 25-20 September 2016
East Lansing, Michigan

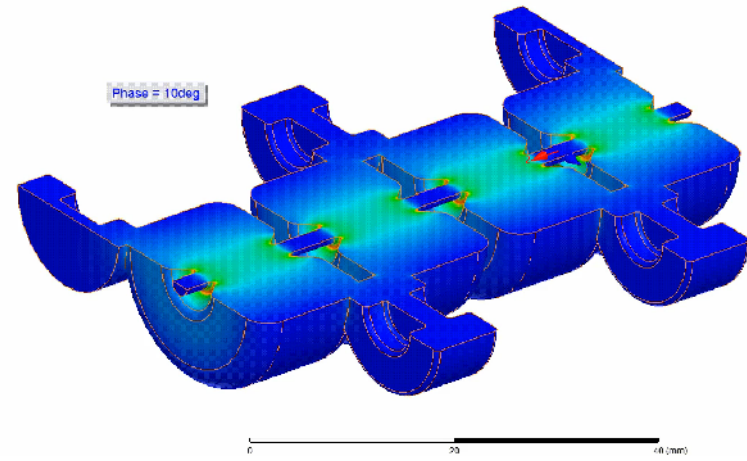
Oral poster MOOP04, 15:15-15:20, Monday, September 26th, 2016

Poster **MOPRC030**, **16:00-17:00**, Monday, September 26th, 2016

Side-Coupled Standing Wave Accelerating Structure

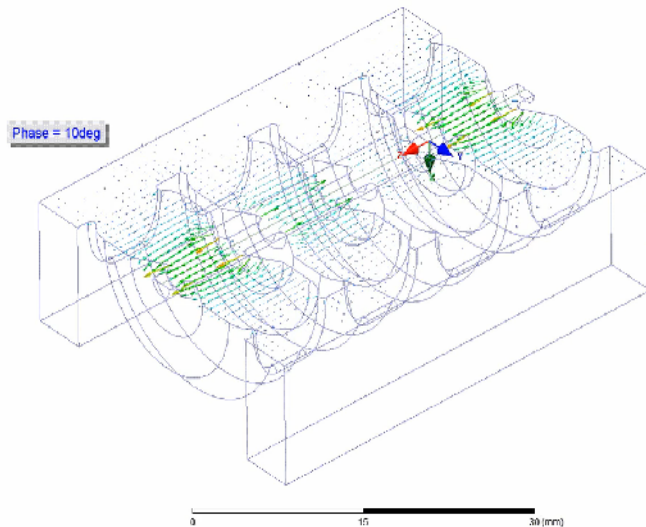


Vector plot of electric fields

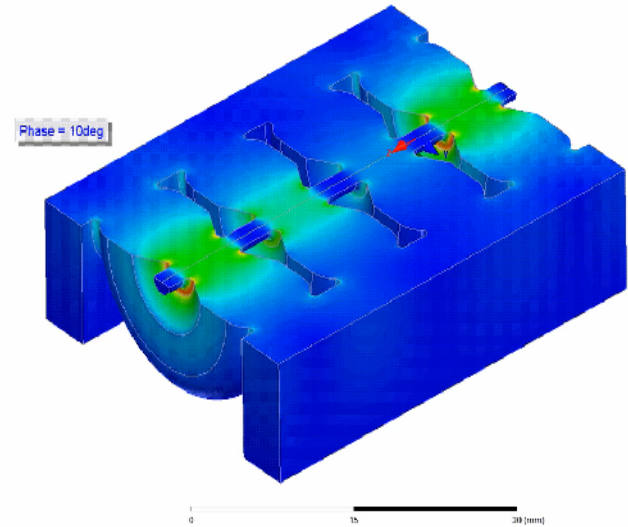


Surface electric fields

New Traveling Wave Accelerating Structure



Vector plot of electric fields



Surface electric fields

Advantages of new structure

Advantages over side-coupled standing wave structures

- No transverse fields kicking the beam off axis.
- The beam loading does not increase wall losses.
- No need in expensive and lossy circulator.

Advantages over parallel coupled standing wave structures

- No transverse fields kicking the beam off axis.
- No need for circulator.

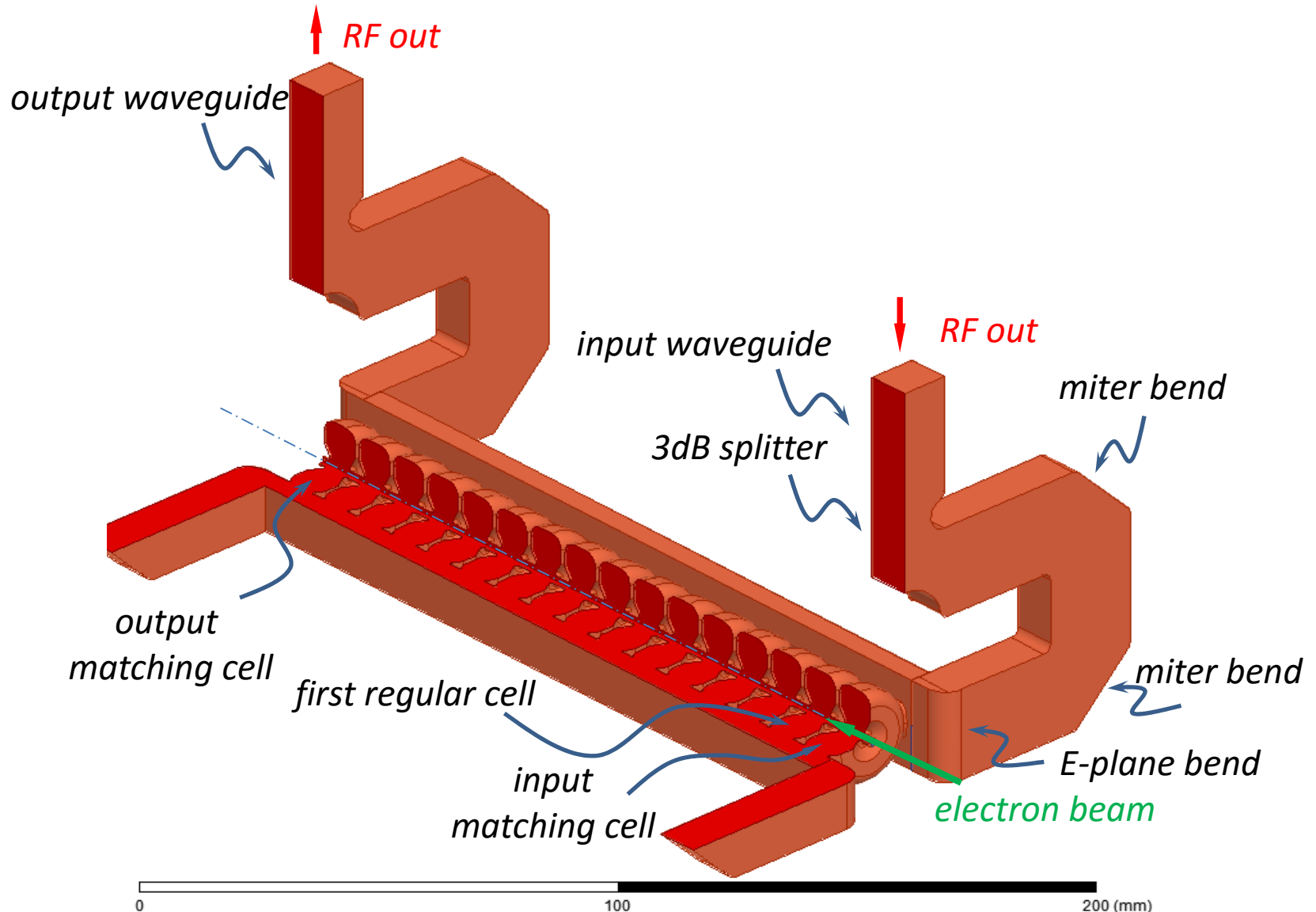
Advantages over on-axis coupled traveling wave structures

- Shunt impedance could be increased with nosecones without affecting coupling between cells.

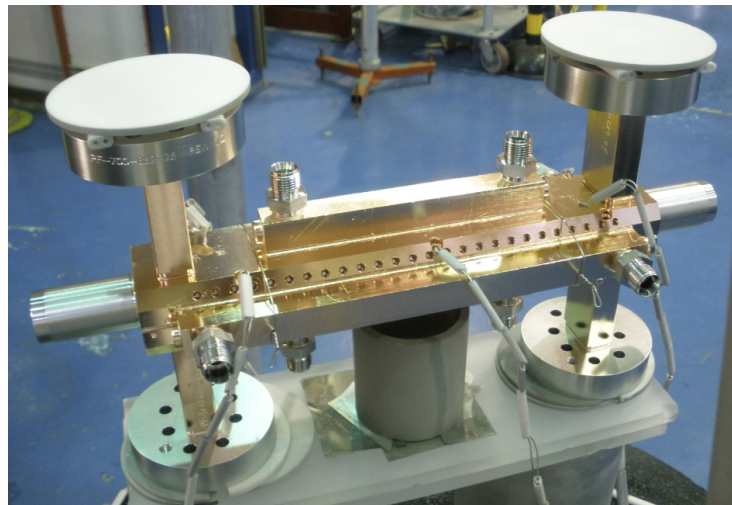
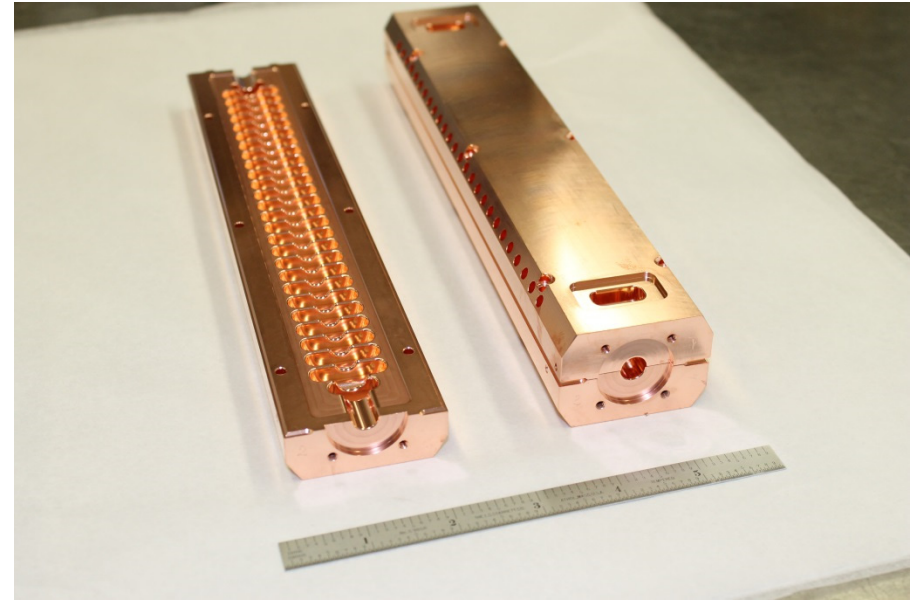
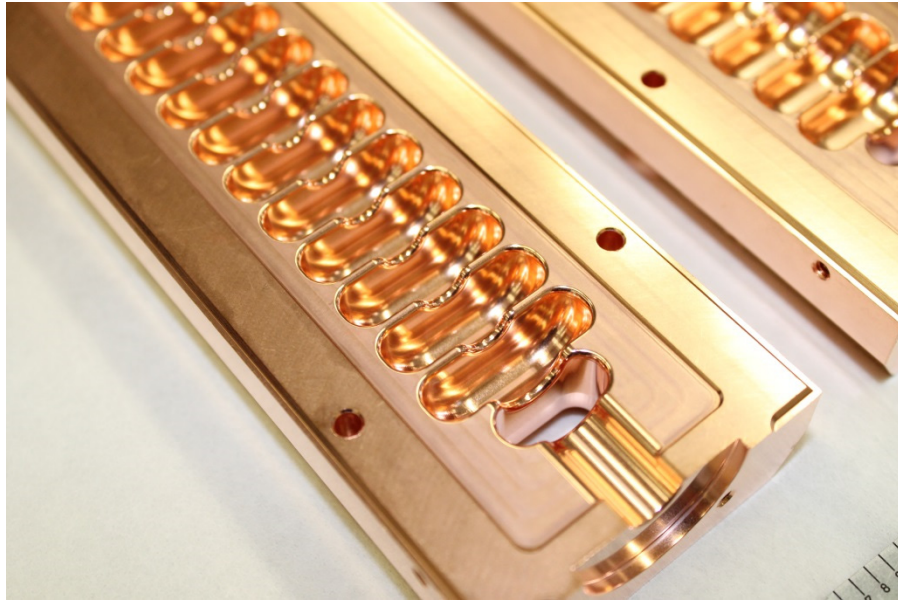
Potential for improved RF breakdowns performance

- *vs. TW and side-coupled SW*: limiting power flow to the breakdown.
- *vs. SW*: For same gradient, smaller stored energy in each cell.

Possible Implementation



Manufacturing: Precision Milling out of Two Halves



“Fabrication and high-gradient testing of an accelerating structure made from milled halves”, [Poster THPLR003 this Thursday](#).

Poster **MOPRC030**, 16:00-17:00, Monday, September 26th, 2016